

Project 1: Eliminating Child Care Deserts in New York State

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1 Introduction

Reliable, affordable child care is an essential part of early childhood development and parental labor-force participation. Yet many New York State communities remain “child care deserts”, which can be defined as **areas where licensed capacity falls well short of local demand**, especially for children ages 0–5. This project investigates how the state could most effectively alleviate deserts under realistic policies constraints. We build an optimization-based framework that:

1. expands capacity at existing facilities
2. sites new facilities of different sizes

while also accounting for age-specific demand, spatial separation requirements, and scale-sensitive expansion costs. Using ZIP code level data on population, employment, income, existing capacity, and candidate locations, we estimate the cost for the New York State government to resolve this issue.

2 Background

Child care provides supervision and early development support for children from two weeks to 12 years old and is essential to enabling parental labor-force participation. In New York, many communities face persistent “child care deserts.”. New York exhibits this pattern across numerous zip codes, making the allocation and expansion of child care capacity a pressing public policy problem.

Defining child care deserts in NY: Let $\mathcal{Z}$ denote the set of zip codes. For $z \in \mathcal{Z}$, let P_z^{0-12} be the number of children aged 0–12 and S_z^{0-12} the number of licensed slots available to them. Let $e_z \in [0, 1]$ be the parent employment rate and $\bar{y}_z$ the average individual income in the area. NYS distinguishes *high-demand* areas by

$$H_z = 1\{e_z \geq 0.60 \text{ or } \bar{y}_z \leq \$60,000\},$$

and sets a demand-adjusted threshold

$$\tau_z = \begin{cases} \frac{1}{2}, & \text{if } H_z = 1 \quad (\text{high-demand}), \\ \frac{1}{3}, & \text{if } H_z = 0 \quad (\text{normal-demand}). \end{cases}$$

Area z is classified as a *child care desert* if $S_z^{0-12} \leq \tau_z P_z^{0-12}$. In addition, NYS emphasizes access for younger children: the available slots for ages 0–5 must satisfy

$$S_z^{0-5} \geq \frac{2}{3} P_z^{0-5} \quad \forall z \in \mathcal{Z}.$$

Policy context: To eliminate deserts, the state considers two different approaches:

- (i) expanding capacity at existing facilities
- (ii) building new facilities of standardized sizes (e.g., small, medium, large)

In Section 3.2: The Problem of Budgeting, expansions are capped for feasibility, meaning no facility may expand beyond 120% of its current capacity. Then in Section 3.3: A more realistic cost view changes the constraint so that there are rising marginal costs as expansions approach the cap.

Siting considerations. To avoid over-concentration, siting decisions must respect a minimum separation between any two facilities, both existing and new, within the same area. The problem suggests a minimum distance of 0.06 miles, introducing a spatial feasibility constraint that interacts with both expansion and newly built facilities.

Data inputs. The project uses the following datasets that are provided:

- (i) licensed facility registry and existing capacity for a given zipcode
- (ii) population by age buckets for a given zipcode
- (iii) employment rates for a given zipcode
- (iv) average individual income for a given zipcode
- (v) a catalog of potential locations for new facilities for a given zipcode

These data jointly determine demand classification, desert status, feasible expansions, and spatial siting constraints.

3 Methods

3.1 Data Processing

Based on the 5 given datasets that were previously mentioned, it was then observed that each dataset contained a different set of ZIP code values for that given ZIP code, represented by:

$$S_i, i \in 1, 2, 3, 4, 5$$

To better visualize the graphic for different ZIP codes, we used a simple script in python to map the union of all ZIP codes mentioned in all 5 datasets, shown mathematically as

$$S_1 \cup S_2 \cup S_3 \cup S_4 \cup S_5$$

We found a total of **2158** number of unique ZIP codes, then we were able to map all these ZIP codes to be visualized as the following:

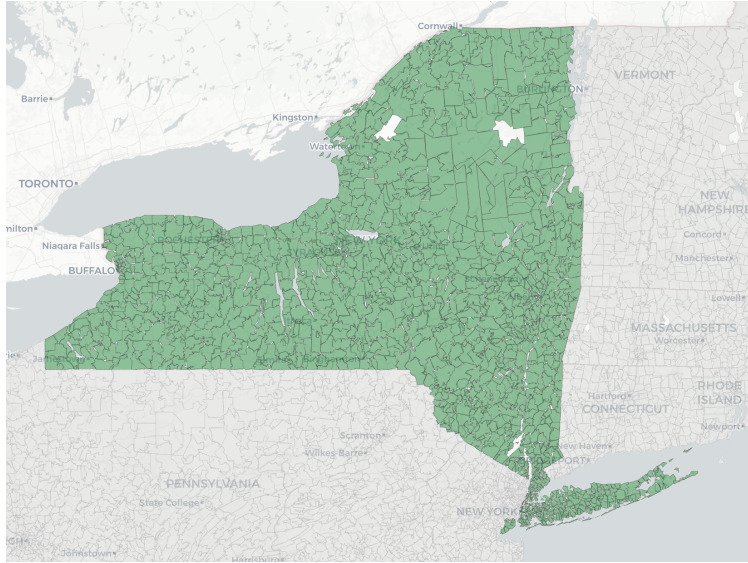


Figure 1: Coverage of union of ZIP codes

While the union of ZIP codes provides full spatial coverage of New York State, not all **2,158** ZIP codes contained complete data across all five datasets. For a given ZIP code i , some datasets were missing values. After filtering out ZIP codes with incomplete data, only **1,023** ZIP codes were found to have values available in all five datasets, while the remaining **1,135** had missing entries. Figure 2 shows the geographic distribution of ZIP codes with complete data.

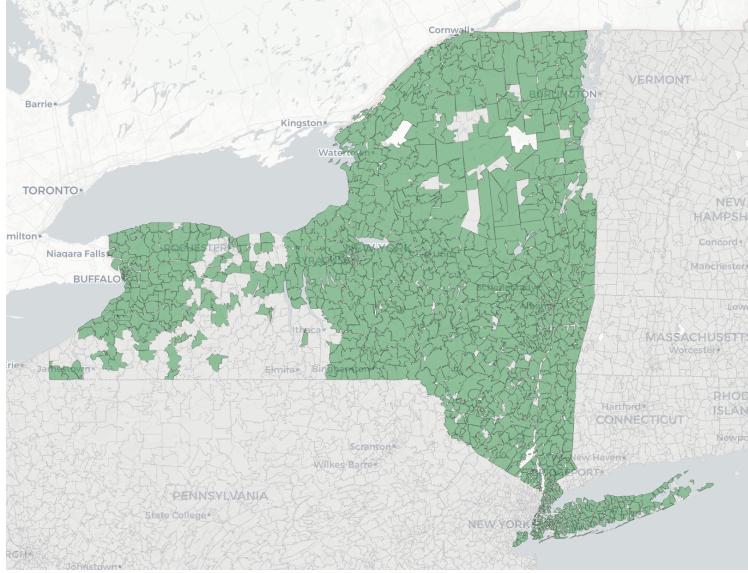


Figure 2: Coverage of ZIP codes with all values

It is evident from Figure 2 that a substantial portion of New York State lacked full data coverage. To address this issue, we attempted to fill the missing ZIP code values using the *U.S. Census Bureau API*¹, which provides demographic and economic data at the ZIP code level. By using this approach, an additional **501** ZIP codes were successfully completed, bringing the total number of ZIP codes with full data coverage to **1,524**. The resulting filled coverage is shown in Figure 3.



Figure 3: Coverage of ZIP codes after retrieving data

Although this dataset is not entirely complete, and alternative methods such as using linear regression to estimate missing values based on neighboring ZIP code areas were considered, the decision was made to use actual data to maintain the realism of the study. Therefore, the **1,524** ZIP codes illustrated in Figure 3 were used as the final dataset for the subsequent optimization process, providing a more reasonable and

¹United States Census Bureau. *Census Data API*. <https://www.census.gov/data/developers/data-sets.html>

representative coverage of New York State compared to using only the 1023 ZIP codes with fully available data.

3.2 The Problem of Budgeting

Assumptions Before constructing the variables and constraints for the optimization process, we first make some assumptions to limit our problem and make it the problem more solvable:

- Demand is static and determined by given data; no changes over time is considered.
- New facilities can be built in any ZIP code, without location restrictions.
- Expansions are limited to an additional 120% of current capacity or a maximum of 500 slots per facility.
- Expansion costs are modeled linearly with respect to the number of slots, with an additional baseline cost triggered when expansions reach 100% of the current facility size.
- Costs of new facilities are fixed depending on their size (Small, Medium, Large).
- Each new slot for children aged 0–5 requires an additional \$100 equipment cost.
- Since there was no defined range value for 0–12 age range in the population dataset, the number of 0–12 children in a given ZIP code is calculated as the population of:

$$(0 \text{ to } 5) + (5 \text{ to } 9) + \frac{3}{5}(10 \text{ to } 14)$$

Sets To model the question, we then create sets to organize the data and define the scope of the problem. Each set represents a collection of related elements that the model needs to consider when making decisions. By defining sets, we can write general equations that automatically apply to every member of a category, without needing to repeat formulas for each one.

- I : set of different ZIP code values within New York State.
- F_i : set of existing facilities located within a given ZIP code area i .
- $S = \{S, M, L\}$: new facility sizes (Small, Medium, Large).
 1. Small = 100 total slots
 2. Medium = 200 total slots
 3. Large = 400 total slots

Parameters Given the sets of data, we can then create parameters based on the data points within each of the sets to connect them together.

- Children_i : population of children aged 0–12 in ZIP code i .
- Infant_i : population of infants aged 0–5 in ZIP code i .
- InfantCap_i : current capacity of infants aged 0–5 in ZIP code i .
- θ_i : coverage threshold in ZIP code i :

$$\theta_i = \begin{cases} \frac{1}{2}, & \text{if ZIP code } i \text{ is high-demand,} \\ \frac{1}{3}, & \text{otherwise.} \end{cases}$$

- Cap_f : current capacity of facility f .

- Size_s : total size of new facility type $s \in S$ (100, 200, 400).
- Slots_s : 0–5 slots of type s (50, 100, 200).
- Cost_s : cost of building a new facility of type s (\$65,000; \$95,000; \$115,000).
- $\beta = \$100$: equipment cost per new 0–5 slot.
- $\alpha = \$200$: per-slot expansion cost.
- $\delta = \$20,000$: baseline cost when an expansion doubles the current size.

Decision Variables These are the decision variables that will impact the cost function and our optimal solution.

- x_f : number of expansion slots at facility f .
- u_f : number of 0–5 slots allocated within expansion x_f .
- $y_{i,s}$: number of new facilities of type s built in ZIP code i .
- $v_{i,s}$: number of 0–5 slots allocated in new facilities of type s in ZIP code i .
- $z_f \in \{0, 1\}$: binary variable indicating whether expansion at facility f reaches or exceeds its current capacity.

Objective Function We minimize the total cost:

$$\min Z = \text{Expand}(x, z) + \text{Facility}(y) + \text{Equip}(u, v).$$

where expansion costs are defined as

- **Expansion Cost Term**

$$\text{Expand}(x, z) = \sum_{f \in F} \text{Expand}_f(x_f, z_f) = \sum_{f \in F} \left[(\delta + 200 \cdot \text{Cap}_f) z_f + \alpha x_f \right].$$

This represents the total cost of expanding existing childcare facilities. The variable x_f represents the number of new slots added to facility f , while the coefficient $\alpha = \$200$ specifies the per-slot expansion cost. The binary variable z_f acts as a trigger that adds an additional baseline cost $\delta + 200 \cdot \text{Cap}_f$ whenever an expansion reaches or exceeds the original capacity of the facility.

- **New Facility Construction Term**

$$\text{Facility}(y) = \sum_{i \in I} \sum_{s \in S} \text{Cost}_s y_{i,s}.$$

This represents the total cost required to construct new childcare facilities. The variable $y_{i,s}$ represents the number of new facilities of size type s (Small, Medium, or Large) built in ZIP code i , while Cost_s corresponds to the fixed cost associated with building each facility of that size.

- **Equipment Cost Term**

$$\text{Equip}(u, v) = \beta \left(\sum_{f \in F} u_f + \sum_{i \in I} \sum_{s \in S} v_{i,s} \right).$$

The coefficient $\beta = \$100$ represents the per-slot equipment and furnishing cost for children aged 0–5. The variables u_f and $v_{i,s}$ measure how many of the expanded or newly built slots are allocated to this age group.

Constraints We can then find the constraints for the optimization based on the different information that can be extracted from the given text:

Expansion limits: Expansions are restricted so that the total capacity after expansion does not exceed either 220% of current capacity or an cap of 500 slots:

$$\text{Cap}_f + x_f \leq \min\{2.2 \cdot \text{Cap}_f, 500\}, \quad \forall f.$$

Coverage requirement (desert elimination): The New York State (NYS) government aims to eliminate child care deserts by increasing the number of child care slots across all regions. The goal is to ensure that no area remains classified as a child care desert.

$$\sum_{f \in F_i} (\text{Cap}_f + x_f) + \sum_{s \in S} \text{Size}_s y_{i,s} \geq \theta_i \cdot \text{Children}_i, \quad \forall i.$$

0–5 coverage requirement: Beyond addressing overall demand, the NYS government ensures that children under the age of 5 have sufficient access to care. This means that the number of available slots for children in this age group must be at least two-thirds of the population of children aged 0-5.

$$\text{InfantCap}_i + \sum_{f \in F_i} u_f + \sum_{s \in S} v_{i,s} \geq \frac{2}{3} \cdot \text{Infant}_i, \quad \forall i.$$

Consistency constraints: These constraints ensure that the number of slots allocated to children aged 0–5 does not exceed the total number of expanded or newly constructed slots. In other words, the allocations u_f and $v_{i,s}$ must remain consistent with the capacity limits set by expansions x_f and newly built facilities $y_{i,s}$:

$$\begin{aligned} 0 &\leq u_f \leq x_f, & \forall f, \\ 0 &\leq v_{i,s} \leq \text{Infant}_s y_{i,s}, & \forall i, s. \end{aligned}$$

Binary trigger for expansion costs: To expand capacity by 100% or more, the state must pay a baseline cost of \$20,000, plus \$200 per existing slot. In other words, when the number of new slots added equals or surpasses the facility's current capacity, the expansion triggers an additional baseline cost of \$20,000 plus \$200 per existing slot.

$$z_f = \begin{cases} 1, & \text{if } x_f \geq \text{Cap}_f, \\ 0, & \text{if } x_f < \text{Cap}_f. \end{cases}$$

Facilities that already exceed their policy capacity limits are excluded from this activation rule, since no further expansion is permitted for them.

Non-zero: All expansion and coverage variables should be strictly positive for all applicable indices.

$$x_f, u_f, y_{i,s}, v_{i,s} > 0 \quad \forall f, i, s$$

3.3 The Problem of Realistic Capacity Expansion and Location

Updated Assumptions To better reflect the complexity of expanding child care facilities and choosing appropriate locations for new ones, we modify our assumptions to be:

- Parameters that were not mentioned in Part 2 remain the same as in Part 1.
- the **Binary trigger for expansion costs** constraint from Part 1 is removed
- Existing facilities can expand by at most 20% of their current capacity.

- Expansion costs rise nonlinearly with scale, divided into three linearized tiers based on the ratio $\frac{x_f}{\text{Cap}_f}$:

$$C_f^{\text{exp}}(x_f) = \begin{cases} (20,000 + 200 \text{Cap}_f) \frac{x_f}{\text{Cap}_f}, & 0 < \frac{x_f}{\text{Cap}_f} \leq 0.10, \\ (20,000 + 400 \text{Cap}_f) \frac{x_f}{\text{Cap}_f}, & 0.10 < \frac{x_f}{\text{Cap}_f} \leq 0.15, \\ (20,000 + 1,000 \text{Cap}_f) \frac{x_f}{\text{Cap}_f}, & 0.15 < \frac{x_f}{\text{Cap}_f} \leq 0.20. \end{cases}$$

- To prevent over-concentration, any two facilities, both new and existing must be at least 0.06 miles apart.
- The distance between two geographic coordinates $(\text{lat}_1, \text{lon}_1)$ and $(\text{lat}_2, \text{lon}_2)$ can be calculated using the **Haversine formula**, which is:

$$d = 2R \cdot \arctan\left(\frac{\sqrt{a}}{\sqrt{1-a}}\right), \quad \text{where} \quad a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\Delta\lambda}{2}\right),$$

with

$$\Delta\phi = \phi_2 - \phi_1, \quad \Delta\lambda = \lambda_2 - \lambda_1, \quad R = 3958.8 \text{ miles.}$$

Updated Decision Variables New location-based binary variables are introduced:

- $y_{i,l,s} \in \{0,1\}$: equals 1 if a new facility of type s is built at potential site l in ZIP code i ; 0 otherwise.
- $v_{i,l,s}$: number of 0–5 slots allocated in the new facility of type s at site l in ZIP i .
- t_f^1, t_f^2, t_f^3 : variables representing expansion tiers (0–10%, 10–15%, 15–20%) for each existing facility f .

We can then defined variables as:

$$y_{i,s} = \sum_l y_{i,l,s}, \quad v_{i,s} = \sum_l v_{i,l,s}.$$

Updated Objective Function The cost function remains in the same form that has the three components:

$$\min Z = \text{Expand}(t) + \text{Facility}(y) + \text{Equip}(u, v).$$

- **Expansion Cost Term**

$$\text{Expand}(t) = \sum_{f \in F} \left[\left(\frac{20,000}{\text{Cap}_f} + 200 \right) t_f^1 + \left(\frac{20,000}{\text{Cap}_f} + 400 \right) t_f^2 + \left(\frac{20,000}{\text{Cap}_f} + 1,000 \right) t_f^3 \right].$$

Here t_f^1, t_f^2, t_f^3 are the expansion slots at facility f in the 0–10%, 10%–15%, and 15%–20% bands (of Cap_f), respectively, and Cap_f is the current capacity. This implements the increasing marginal cost by band while spreading the \$20,000 baseline over Cap_f slots in each tier.

- **Facility(y)** and **Equip(u, v)** remain the same as defined in Part 3.2.

Updated Constraints

Expansion capacity tiers: Each facility's expansion is partitioned into tiered slots:

$$x_f = t_f^1 + t_f^2 + t_f^3 \quad t_f^1 \leq 0.10 \text{Cap}_f, \quad t_f^2 \leq 0.05 \text{Cap}_f, \quad t_f^3 \leq 0.05 \text{Cap}_f.$$

Distance feasibility among new sites: No two selected potential sites within the same ZIP may violate the 0.06-mile spacing rule:

$$\sum_{s \in S} y_{i,a,s} + \sum_{s \in S} y_{i,b,s} \leq 1, \quad \forall (a, b): d_i(a, b) < 0.06.$$

Distance feasibility between new and existing facilities: Potential sites too close to existing facilities are excluded:

$$\sum_{s \in S} y_{i,l,s} \leq 1, \quad \forall (l, f): d_i(l, f) < 0.06.$$

One facility per site: Each potential site may only have one facility built:

$$\sum_{s \in S} y_{i,l,s} \leq 1, \quad \forall i, l.$$

3.4 The Problem of Fairness

Motivation. While Parts 3.2 and 3.3 identify minimum-cost strategies to eliminate deserts, policymakers also seek to ensure that access improvements are distributed equitably across regions. To avoid large disparities between ZIP codes, we introduce a **fairness formulation** that maximizes statewide social coverage subject to a fixed budget of \$100,000,000, while keeping regional coverage rates within a 10 percentage point band.

Coverage definitions. For each ZIP code i , the total available childcare slots by age group—including existing, expanded, and newly constructed capacity—are computed as:

$$S_i^{0-5} = \text{InfantCap}_i + \sum_{f \in F_i} u_f + \sum_{s \in S} v_{i,s}, \quad (1)$$

$$S_i^{5-12} = \sum_{f \in F_i} (\text{Cap}_f - u_f + x_f) + \sum_{s \in S} (\text{Size}_s - v_{i,s}) y_{i,s}, \quad (2)$$

The corresponding coverage ratios use the population parameters already defined in Part 3.2:

$$R_i^{0-5} = \frac{S_i^{0-5}}{\text{Infant}_i}, \quad R_i^{5-12} = \frac{S_i^{5-12}}{\text{Children}_i - \text{Infant}_i}, \quad R_i^{0-12} = \frac{S_i^{0-5} + S_i^{5-12}}{\text{Children}_i}.$$

Objective (social coverage index). Younger children remain the policy priority, so their coverage receives greater weight. The statewide objective maximizes a weighted social coverage index:

$$\max Z = \sum_{i \in I} \left[\frac{2}{3} R_i^{0-5} + \frac{1}{3} R_i^{5-12} \right]. \quad (3)$$

Because all population parameters are constants, (3) is linear in the decision variables $x_f, u_f, y_{i,s}, v_{i,s}$.

Fairness constraints. To ensure equitable access, no two ZIP codes' overall (0–12) coverage rates should differ by more than 10 percentage points:

$$|R_i^{0-12} - R_j^{0-12}| \leq 0.10, \quad \forall i, j \in I. \quad (4)$$

For computational efficiency, this can be approximated by comparing each ZIP to a statewide average coverage level $\bar{R}$:

$$|R_i^{0-12} - \bar{R}| \leq 0.10, \quad \bar{R} = \frac{1}{|I|} \sum_{i \in I} R_i^{0-12}. \quad (5)$$

Budget constraint. Unlike the unconstrained cost-minimization problems in Parts 3.2–3.3, this fairness formulation operates under a fixed statewide budget $B = \$100,000,000$:

$$\text{Expand}(x, z) + \text{Facility}(y) + \text{Equip}(u, v) \leq B.$$

This ensures that total spending on expansions, construction, and equipment remains within the allocated funding.

4 Results

4.1 Part 1:

Based on the Linear Program defined in Part 3.2, a gurobi script was written in python to calculate the optimal value (minimum cost), which gave a optimal value of:

\$324,386,100

4.2 Part 2:

Based on the Linear Program defined in Part 3.3, a gurobi script was written in python to calculate the optimal value (minimum cost), which gave a optimal value of:

\$501,493,891

Note: During the optimization process, an outlier facility was identified within ZIP code **11225**, facility ID **707743**, named *Friends of Crown Heights #11 Educational Center* in Brooklyn. This facility reported a total capacity of zero, to ensure consistency and prevent divide-by-zero or infeasible cases during optimization, it was removed from both the expansion and location consideration datasets. The metadata is shown below:

```
"707743": {  
  "facility_name": "Friends of Crown Heights #11 Educational Center",  
  ...  
  "total_capacity": 0,  
  "latitude": 40.667762,  
  "longitude": -73.957164  
}
```

4.3 Graph Interpretation and Analysis

Newly added and expanded slots Figures 4 and 5 visualize how additional childcare capacity was distributed geographically across ZIP codes.

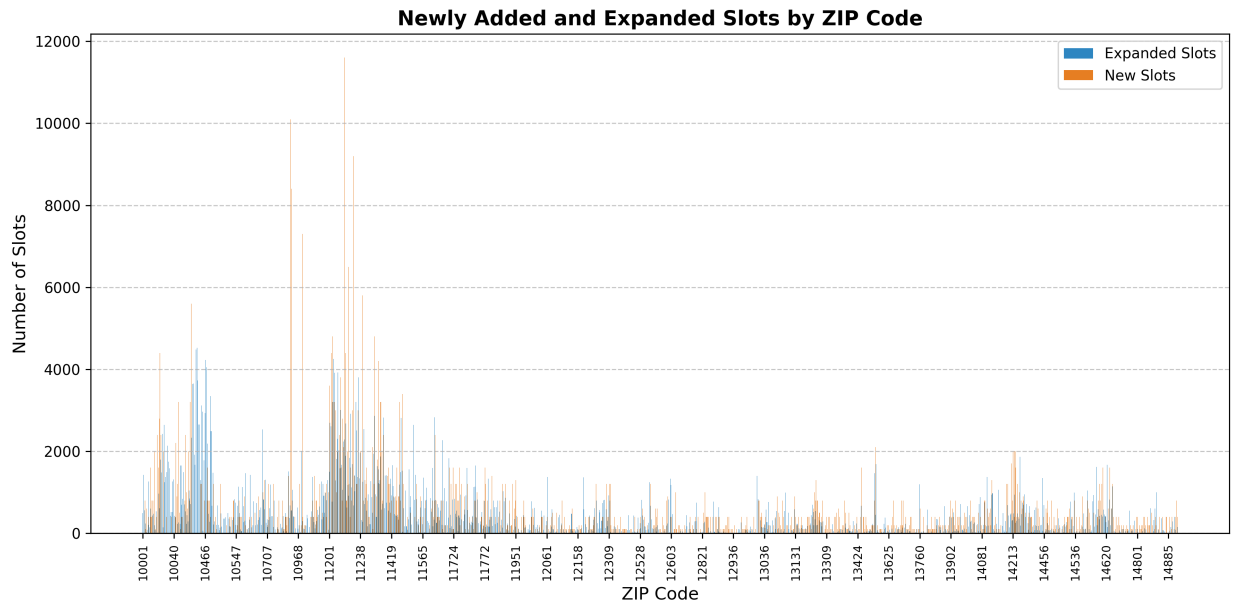


Figure 4: Newly Added and Expanded slots - Part 1

In **Part 1**, where expansions were unconstrained, the majority of capacity growth was achieved through expansions. Especially concentrated in urban ZIP codes with a higher density of existing facilities. New builds were fewer and primarily located in underserved suburban and rural regions lacking sufficient baseline capacity.

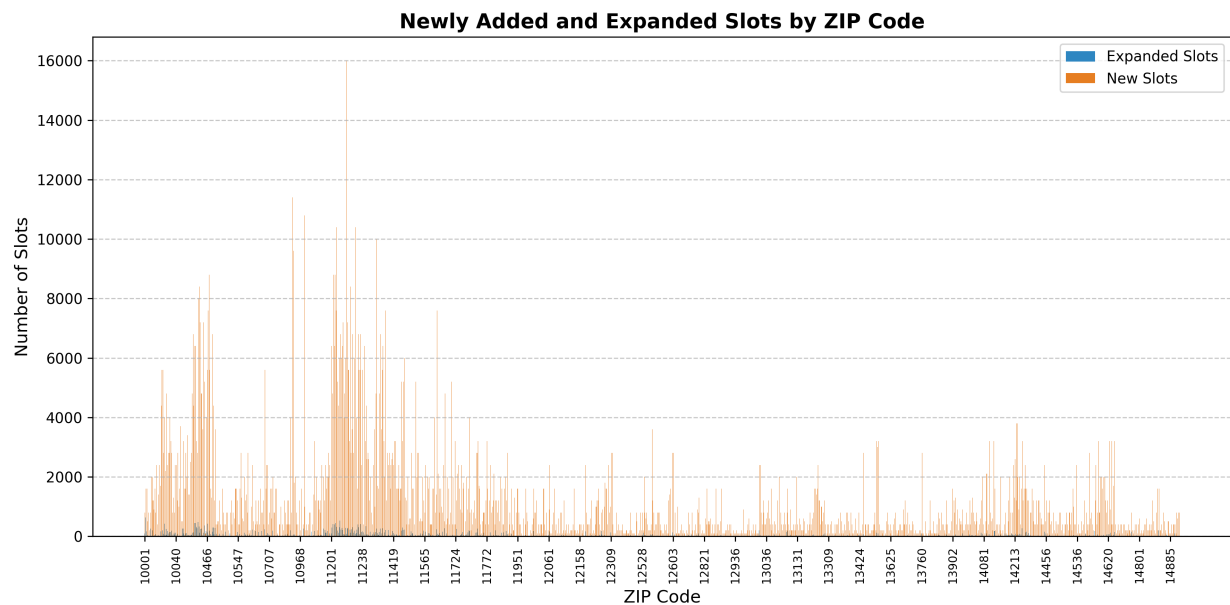
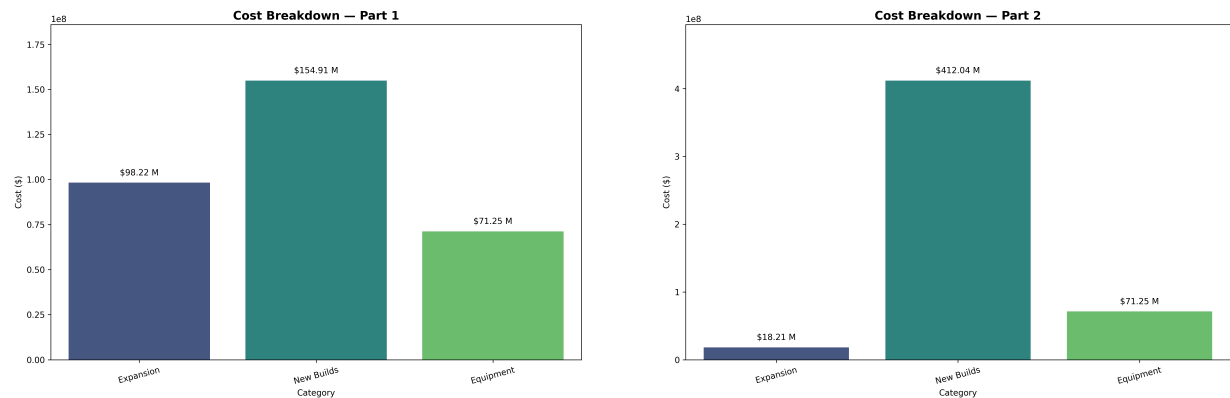


Figure 5: Newly Added and Expanded slots - Part 2

In **Part 2**, the introduction of spatial and physical constraints shifts the balance decisively toward new facility construction. Many ZIP codes that relied heavily on expansion in Part 1 now exhibit minimal or no expansion activity, as existing centers quickly reach their growth limits. Consequently, new expansions becomes the main way of meeting unmet childcare demand, especially in high-density or constrained areas.

Cost Breakdown: Figures 6a and 6b shows the total spending composition across the three cost categories: **Expansions**, **New Builds**, and **Equipment**.



(a) Cost Breakdown Part 1

(b) Cost Breakdown Part 2

Figure 6: Comparison of total cost composition between Part 1 and Part 2.

In Part 1, the costs are distributed more evenly between expansion and new builds, suggesting that

unconstrained conditions allow the model to rely more on expanding existing centers rather than building new ones. In Part 2, the introduction of spatial and practical constraints significantly increases the relative cost of meeting demand, as expansions are capped and new builds become the dominant cost driver. Equipment costs remain relatively proportional, scaling with the number of 0–5 slots added. The total cost increases from roughly $320M$ to $512M$ shows that realistic constraints make new builds the main cost burden.

Average Expansion per ZIP Code Figures 7a and 7b shows the average expansion number within each ZIP code. Each bar corresponds to the number of ZIP codes that experienced expansions within a certain range of additional slots.

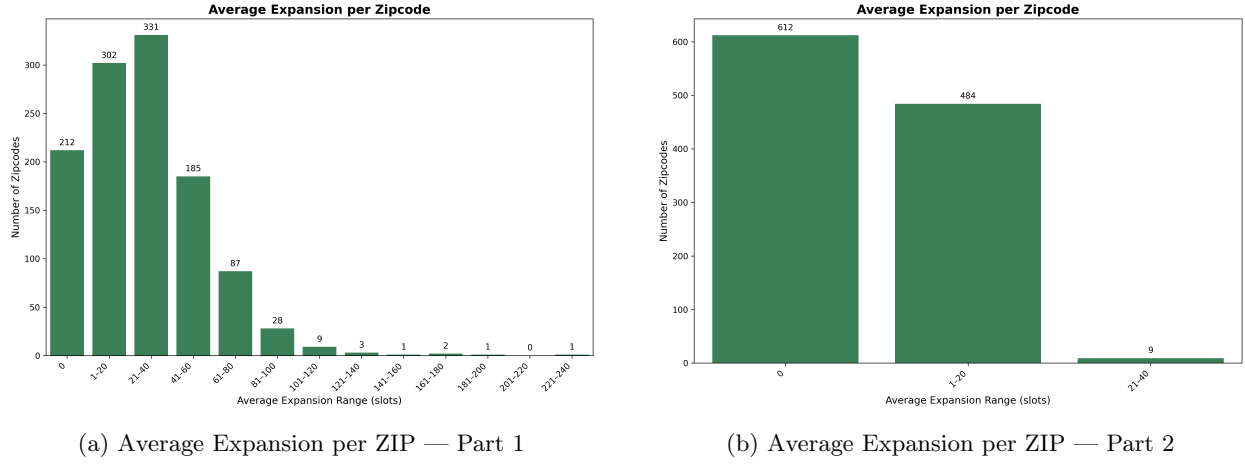


Figure 7: Comparison of Average Expansion per ZIP between optimization scenarios.

In Part 1, the distribution is wider, indicating that a substantial number of ZIP codes experienced moderate to large expansions. In contrast, Part 2’s distribution is sharply concentrated in the 0–20 slot range, with most ZIP codes experiencing minimal or no expansions. This reflects how the 20% cap and tiered cost model restrict expansion opportunities, shifting most of the coverage gains to new facility construction rather than existing site growth. As a result, new builds replace large expansions as the main strategy for capacity growth.

0–5 Expansion Slots per Facility Figures 8a and 8b show how expansions targeted specifically at 0–5 year-old childcare slots are distributed across individual facilities.

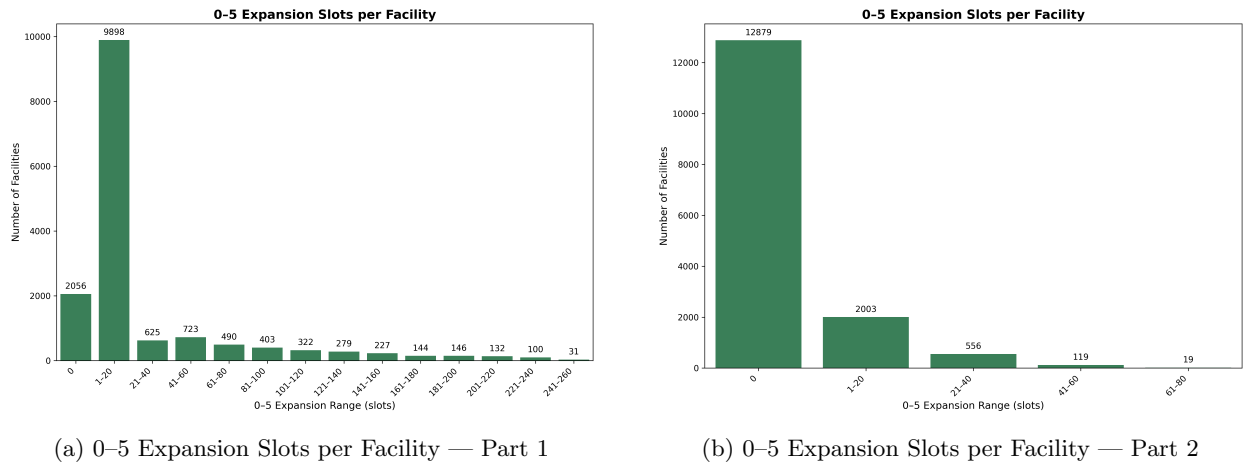


Figure 8: Comparison of 0–5 Expansion Slots per Facility between optimization scenarios.

In both Part 1 and Part 2, the results are highly skewed. Most facilities add only a small number (1–20) of 0–5 slots, while some facilities make larger additions. From Part 1 to Part 2, the mass of the distribution shifts toward zero, meaning that realistic physical constraints further reduce the number of facilities capable of meaningful 0–5 expansions. This has important implications for compliance with the two-thirds coverage rule for children under 5 years old indicating that expansions alone can’t meet that coverage goal so new centers must fill the remaining gap.

5 Conclusion

5.1 Analysis

Our optimization results demonstrate that eliminating child care deserts across New York State is operationally feasible under the stated assumptions and constraints. Using a combination of targeted expansions at existing providers and the construction of standardized new facilities, the model identifies cost-minimizing strategies that satisfy statewide coverage requirements.

Model Outcomes. In the baseline Part 1, where expansions were unconstrained, the optimizer achieved statewide coverage at a total estimated cost of **\$324,386,100**. Most capacity gains occurred through expansions concentrated in urban ZIP codes, while new builds primarily filled gaps in underserved suburban and rural regions. When realistic spatial and physical constraints were introduced in Part 2, the minimum feasible cost rose to **\$501,493,891**. The direction of this change aligns with intuition: once low-cost expansion options were exhausted, the optimizer shifted toward new construction as the dominant strategy.

Data Treatment and Preprocessing. The modeling choices around data completeness and cleaning were critical to ensuring reliable outcomes. Constructing the union of ZIP codes, filling missing demographic fields with Census data, and removing outliers improved representativeness and prevented computational issues.

Comparative Patterns.

- In the baseline scenario, **expansions** account for most added slots and spending.
- In the realistic scenario, **new builds** dominate both capacity growth and total cost.

This shift illustrates that inexpensive incremental expansions are quickly constrained by physical and geographic limits, requiring substantial investment in new construction to close remaining gaps.

Overall, the model provides a quantitative foundation for statewide childcare expansion planning. It demonstrates that meeting equity and coverage goals is feasible—but only with realistic budgeting, expanded candidate site inventories, and a clear focus on the 0–5 age group, where demand is both highest and most difficult to meet through simple expansions.

5.2 Reflection

Our current model provides a solid foundation for addressing the problem, but several areas for improvement still exist. We acknowledge these limitations to show that they were considered, but given the time and resource constraints of this project, they were not fully explored. Addressing these in future iterations would improve the precision and realism of the policy implications even further.

1. Data Completeness. The biggest shortcoming and incompleteness comes from the set of ZIP codes included in the available datasets. However, since some ZIP codes remain missing due to incomplete Census or facility reporting data. This leads to uneven geographic coverage, particularly in rural or newly reclassified areas.

2. Cross-ZIP Distance Constraints. In Part 2, the distance restriction of 0.06 miles was only enforced within the boundaries of a single ZIP code. In practice, facilities located near ZIP borders could violate this rule if two nearby sites happen to lie on opposite sides of a ZIP boundary.

3. Dynamic Nature of Demand and Time. The model currently treats both population and demand as static within each ZIP code. In reality, these factors evolve continuously with migration trends, commuting patterns, and shifts in local employment or school access. Families frequently cross ZIP boundaries when choosing care, making demand highly dynamic rather than geographically fixed. Moreover, construction and licensing timelines, as well as inflation and demographic change, influence when new capacity becomes available.

6 Code

Code without the datasets was uploaded to a GitHub repository: https://github.com/cctofu/childcare_desert
Code with the full dataset and all files were compressed and submitted to Gradescope